

Data Mining Numerical Problems & Solutions

CT 72502 (Elective I) · BEI / BEX IV-I · TU, IOE · Chapter 5: Cluster Analysis

Ch 5 – Cluster Analysis – Numerical Problems & Solutions

3 methods · 11 PYQ instances · every distance recomputed. K-means appears in 7 papers and hierarchical dendrograms in 4, so these two methods alone are worth around 60 marks across the archive.

Methods

- **M1** K-means 78 Bh, 74 Ash, 75 Ash, 71 Shr, 76 Ch, 81 Ba
 - **M2** Agglomerative hierarchical clustering + dendrogram 80 Bh, 80 Ba, 76 Ch
 - **M3** DBSCAN point classification *practice*
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M1. K-means

Method

1. State K and the initial centroids. **Always write down which points you seeded with** — the answer depends on it and the examiner needs to follow you.
 2. Build a distance table: every point against every centroid.
 3. Assign each point to its nearest centroid.
 4. Recompute each centroid as the **mean** of its members.
 5. Repeat until the assignment does not change. **That repetition is the convergence proof.**
- You may compare *squared* distances to avoid square roots — the ordering is identical. Say that you are doing it.

Problems 1.1 and 1.2 2+6 2+8

78 Bh and 74 Ash print the same 6 instances (A/B vs X/Y column names only). $K = 2$.

Given

$p_1(1, 2), p_2(2.5, 1), p_3(3.5, 1.5), p_4(4, 1), p_5(3.5, 2.5), p_6(5, 3)$.

Seeds

Take the two **most widely separated** points, $c_1 = p_1(1, 2)$ and $c_2 = p_6(5, 3)$. (A standard, defensible heuristic; state it.)

Iteration 1

Point	d to $c_1(1, 2)$	d to $c_2(5, 3)$	Cluster
$p_1(1, 2)$	0	4.123	C1
$p_2(2.5, 1)$	1.803	3.202	C1
$p_3(3.5, 1.5)$	2.550	2.121	C2
$p_4(4, 1)$	3.162	2.236	C2
$p_5(3.5, 2.5)$	2.550	1.581	C2
$p_6(5, 3)$	4.123	0	C2

- $C1 = \{p_1, p_2\} \rightarrow$ new centroid $(\frac{1+2.5}{2}, \frac{2+1}{2}) = (1.75, 1.5)$
- $C2 = \{p_3, p_4, p_5, p_6\} \rightarrow$ new centroid $(\frac{3.5+4+3.5+5}{4}, \frac{1.5+1+2.5+3}{4}) = (4, 2)$

Iteration 2

Point	d to $(1.75, 1.5)$	d to $(4, 2)$	Cluster
p_1	0.901	3.000	C1
p_2	0.901	1.803	C1
p_3	1.750	0.707	C2
p_4	2.305	1.000	C2
p_5	2.016	0.707	C2
p_6	3.580	1.414	C2

- **Assignment unchanged $\rightarrow$ converged.**

Final answer

- **Cluster 1** = $\{(1, 2), (2.5, 1)\}$, centroid $(1.75, 1.5)$
- **Cluster 2** = $\{(3.5, 1.5), (4, 1), (3.5, 2.5), (5, 3)\}$, centroid $(4, 2)$
- Converged in **2 iterations**.

Seed sensitivity — worth a line, since the question asks for limitations. Seeding instead with the *first* two points, $c_1 = p_1(1, 2)$ and $c_2 = p_2(2.5, 1)$, gives $C1 = \{p_1\}$ and $C2 = \{p_2, \dots, p_6\}$, which also converges but leaves a **degenerate 1-point cluster**. Same data, same K , **different answer** — this is exactly the “sensitive to initial centroids / converges to a local optimum” limitation. verified/added

Problem 1.3 4+6

75 Ash 6 rows: $(1, 2), (2.5, 4.5), (4, 6), (3.5, 4), (4, 5.5), (3, 6)$. $K = 2$.

- Seeds: $c_1 = (1, 2)$ (clearly isolated), $c_2 = (4, 6)$.
- **Iteration 1:** $(1, 2) \rightarrow C1$. All of $(2.5, 4.5), (4, 6), (3.5, 4), (4, 5.5), (3, 6)$ are nearer $(4, 6) \rightarrow C2$.
 - Eg $(2.5, 4.5)$: d to $c_1 = \sqrt{1.5^2 + 2.5^2} = 2.92$; to $c_2 = \sqrt{1.5^2 + 1.5^2} = 2.12 \rightarrow C2$.
 - Eg $(3.5, 4)$: d to $c_1 = \sqrt{2.5^2 + 2^2} = 3.20$; to $c_2 = \sqrt{0.5^2 + 2^2} = 2.06 \rightarrow C2$.

- New centroids: $C1 = (1, 2)$; $C2 = \left(\frac{2.5+4+3.5+4+3}{5}, \frac{4.5+6+4+5.5+6}{5}\right) = (3.4, 5.2)$
- **Iteration 2:** every $C2$ member stays nearer $(3.4, 5.2)$ than $(1, 2) \rightarrow$ unchanged $\rightarrow$ **converged**.

Final answer

$C1 = \{(1, 2)\}$, $C2 = \{(2.5, 4.5), (4, 6), (3.5, 4), (4, 5.5), (3, 6)\}$, centroid $(3.4, 5.2)$.

- **Note:** $(1, 2)$ is a genuine outlier here, so the 1-point cluster is the honest answer, not a seeding artefact. Say so, and add that **K-medoids** or **DBSCAN** would treat it as noise.

Problem 1.4 10

71 Shr 7 rows: $(1,1), (1.5,2), (3,4), (5,7), (3.5,5), (4.5,5), (3.5,4.5)$. $K = 2$.

- Seeds $c_1 = (1, 1)$, $c_2 = (5, 7)$.
- **Iteration 1:** $(1, 1)$ and $(1.5, 2) \rightarrow C1$; the rest $\rightarrow C2$.
 - $(3, 4)$: d to $c_1 = \sqrt{4+9} = 3.61$; to $c_2 = \sqrt{4+9} = 3.61$ — **an exact tie**. Break it toward $C2$ (state your rule).
- Centroids: $C1 = (1.25, 1.5)$; $C2 = \left(\frac{3+5+3.5+4.5+3.5}{5}, \frac{4+7+5+5+4.5}{5}\right) = (3.9, 5.1)$
- **Iteration 2:** assignments unchanged $\rightarrow$ converged.

Final answer

$C1 = \{(1, 1), (1.5, 2)\}$ centroid $(1.25, 1.5)$; $C2 = \{(3, 4), (5, 7), (3.5, 5), (4.5, 5), (3.5, 4.5)\}$ centroid $(3.9, 5.1)$.

Problem 1.5 8

76 Ch 10 points in matrix X , $K = 3$, initial centres red $(6.2, 3.2)$, green $(6.6, 3.7)$, blue $(6.5, 3.0)$. Give the final centres and comment on the number of iterations.

Given

$p_1(5.9, 3.2)$ $p_2(4.6, 2.9)$ $p_3(6.2, 2.8)$ $p_4(4.7, 3.2)$ $p_5(5.5, 4.2)$ $p_6(5.0, 3.0)$ $p_7(4.9, 3.1)$ $p_8(6.7, 3.1)$ $p_9(5.1, 3.8)$ $p_{10}(6.0, 3.0)$

Iteration 1

(assign to nearest of the 3 given centres)

- **red** $\{p_1, p_2, p_4, p_6, p_7, p_9, p_{10}\}$ **green** $\{p_5\}$ **blue** $\{p_3, p_8\}$
- New centres: red $(5.171, 3.171)$, green $(5.5, 4.2)$, blue $(6.45, 2.95)$

Iteration 2

- **red** $\{p_2, p_4, p_6, p_7\}$ **green** $\{p_5, p_9\}$ **blue** $\{p_1, p_3, p_8, p_{10}\}$
- New centres: red $(4.8, 3.05)$, green $(5.3, 4.0)$, blue $(6.2, 3.025)$

Iteration 3

- Every point stays where it was $\rightarrow$ **converged**.

Final answer

Cluster	Members	Final centre
red	p_2, p_4, p_6, p_7	(4.80, 3.05)
green	p_5, p_9	(5.30, 4.00)
blue	p_1, p_3, p_8, p_{10}	(6.20, 3.025)

- **Comment on iterations:** assignments changed at iterations 1 and 2 and were stable at iteration 3, so **convergence took 3 iterations (2 real updates + 1 confirming pass)**.
- Worth adding: the 3 supplied centres were all bunched on the right of the data ($x \approx 6.2$ – 6.6), yet K-means still pulled them apart into a sensible left / middle / right split. Good illustration that it converges quickly even from a poor start.

Problem 1.6 – the defective question 5+3

81 Ba “Use K-means clustering to cluster the following given data for $K = 2$ with Euclidean distance matrix. List down the demerits of this algorithm.”

The paper prints NO data table for this question — it jumps straight from the sentence to Q5. Verified against the original scan and recorded in the PYQ archive.

In the exam: answer the 3-mark demerits part in full (§5.2), write the 5 steps of the algorithm, and demonstrate on any small dataset of your own, stating that you assumed one. Use **Problem 1.1's** 6 points — it is the dataset this paper's sister questions use. **verified/added**

M2. Agglomerative hierarchical clustering

Method

1. Start with each point as its own cluster; write the distance matrix.
2. Find the **smallest** entry → merge those 2 clusters. **Record the merge height** — that value is the dendrogram bar.
3. Update the matrix using the linkage rule: **single** = MIN of the two rows, **complete** = MAX.
4. Repeat until one cluster remains, then draw the dendrogram with merges at their heights.
 - **Merge heights never decrease.** If yours does, you have made an arithmetic slip.
 - n points always give exactly $n - 1$ merges.

Problem 2.1 2+7

80 Bh 6×6 distance matrix for $p_1 - p_6$; draw the dendrogram. Distances are given directly, so no coordinate work. Use **single link**.

Merge sequence

Step	Merge	Height	Updated distances (MIN)
1	p_3, p_6	0.11	to p_1 0.22, p_2 0.15, p_4 0.15, p_5 0.28
2	p_2, p_5	0.14	to p_1 0.24, p_4 0.20, $\{p_3 p_6\}$ 0.15
3	$\{p_3 p_6\}, p_4$	0.15	to p_1 0.22, $\{p_2 p_5\}$ 0.15
4	$\{p_3 p_4 p_6\}, \{p_2 p_5\}$	0.15	to p_1 0.22
5	all, + p_1	0.22	—

Final answer

Dendrogram heights: **0.11, 0.14, 0.15, 0.15, 0.22.**

- Reading it: $\{p_3, p_6\}$ is the tightest pair; p_1 is the most isolated point, joining last.
- Cutting between 0.15 and 0.22 gives **2 clusters**: $\{p_1\}$ and $\{p_2, p_3, p_4, p_5, p_6\}$.

Problem 2.2 8

80 Ba 6 points given as *coordinates*; use **complete linkage** and draw the dendrogram.

Step 1: Euclidean distance matrix

	p_1	p_2	p_3	p_4	p_5	p_6
$p_1(0.40, 0.53)$						
$p_2(0.22, 0.38)$	0.234					
$p_3(0.35, 0.32)$	0.216	0.143				
$p_4(0.26, 0.19)$	0.368	0.194	0.15			
$p_5(0.08, 0.41)$	0.342	0.143	0.28	0.15		
$p_6(0.45, 0.30)$	0.235	0.244	0.10	0.15	0.15	

Step 2: merge sequence (MAX update)

Step	Merge	Height	Updated distances (MAX)
1	p_3, p_6	0.102	to p_1 0.235, p_2 0.244, p_4 0.219, p_5 0.386
2	p_2, p_5	0.143	to p_1 0.342, p_4 0.284, $\{p_3 p_6\}$ 0.386
3	$\{p_3 p_6\}, p_4$	0.219	to p_1 0.368, $\{p_2 p_5\}$ 0.386
4	$p_1, \{p_2 p_5\}$	0.342	to $\{p_3 p_4 p_6\}$ 0.386
5	the two remaining	0.386	—

Final answer

Heights **0.102, 0.143, 0.219, 0.342, 0.386**; cutting below 0.386 gives $\{p_1, p_2, p_5\}$ and $\{p_3, p_4, p_6\}$.

Problem 2.3 4+4

76 Ch Same 6-object distance matrix (A–F), asked for **both** single-link and complete-link dendrograms. **This is the ideal question for showing you understand the difference.**

(a) Single link (MIN)

	Merge	Height
1	A, B	0.12
2	C, D	0.14
3	{AB}, {CD}	0.16
4	{ABCD}, F	0.20
5	all, + E	0.28

- Heights: **0.12, 0.14, 0.16, 0.20, 0.28**
- Everything merges early and at similar heights → **chaining**.

(b) Complete link (MAX)

	Merge	Height
1	A, B	0.12
2	C, D	0.14
3	{AB}, F	0.61
4	{CD}, E	0.70
5	the two groups	0.93

- Heights: **0.12, 0.14, 0.61, 0.70, 0.93**
- Final 2 clusters: **{A,B,F}** and **{C,D,E}** – a balanced split.

The comparison is the marks. The first 2 merges are identical, then the methods diverge completely: single link strings everything onto the A–B–C–D chain and leaves E as a straggler, while complete link produces two clean balanced groups. **Single link** chases the nearest neighbour, so it follows chains and elongated shapes. **Complete link** is held back by the worst pair, so it prefers compact globular clusters. verified/added

M3. DBSCAN point classification**Method**

1. For each point, count how many points lie within ε (**include the point itself**).
2. Count $\geq$ MinPts → **core**.
3. Otherwise, within ε of a core point → **border**.
4. Otherwise → **noise**.
5. Join core points within ε of each other into clusters; attach the border points.

Practice

No PYQ sets a DBSCAN numerical, but the algorithm is asked in 8 papers and a worked classification makes the definitions concrete. verified/added

Problem

Points $A(1, 1)$, $B(1, 2)$, $C(2, 1)$, $D(8, 8)$, $E(8, 9)$, $F(25, 25)$. Take $\varepsilon = 2$, MinPts = 3.

Step 1: neighbourhood counts

	Point	Points within $\varepsilon = 2$	Count	Type
(including the point itself)	$A(1, 1)$	A, B (1.00), C (1.00)	3	core
	$B(1, 2)$	B, A (1.00), C (1.41)	3	core
	$C(2, 1)$	C, A (1.00), B (1.41)	3	core
	$D(8, 8)$	D, E (1.00)	2	border
	$E(8, 9)$	E, D (1.00)	2	border
	$F(25, 25)$	F only	1	noise

Step 2: form clusters

- A, B, C are all core and all within ε of each other → **Cluster 1** = $\{A, B, C\}$.
- D and E have only 2 neighbours each, and **neither is within ε of any core point** → they are **not border points** after all, **they are noise**.
- F is isolated → **noise**.

Final answer

- **1 cluster**: $\{A, B, C\}$. **Noise**: D, E, F .
- **The trap**: D and E look like a cluster to the eye, but with MinPts = 3 a pair can never form one. **A border point must be within ε of an actual core point** – being near a non-core point is not enough.
- Lower MinPts to 2 and $\{D, E\}$ becomes a second cluster. **Quote your parameters with the answer.**

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

Short	Means	Short	Means
DM	data mining	DW	data warehouse
DB	database	DBMS	database management system
KDD	knowledge discovery in databases	DSS	decision support system
OLTP	on-line transaction processing	OLAP	on-line analytical processing
ROLAP	relational OLAP	MOLAP	multidimensional OLAP
HOLAP	hybrid OLAP	ETL	extract, transform, load
BI	business intelligence	ER	entity-relationship
attr(s)	attribute(s)	dim(s)	dimension(s)
multidim	multidimensional	txn(s)	transaction(s)
info	information	ops	operations
repo	repository	freq	frequent / frequency
config	configuration	no.	number
PCA	principal component analysis	SVD	singular value decomposition
SMC	simple matching coefficient	KNN	<i>k</i> -nearest neighbour
ANN	artificial neural network	POS	point of sale
FK	foreign key	UI	user interface
GIGO	garbage in, garbage out	w/, w/o	with, without
rm	remove	vs	versus
Eg	example / for example	+Fig	draw a figure
→	leads to / therefore	+	and

Exam-paper month codes (used in the year tags): Ba = Baishakh · Jth = Jestha · Asa = Ashar · Shr = Shrawan · Bh = Bhadra · Ash = Ashwin · Ka = Kartik · Mng = Mangsir · Po = Poush · Ma = Magh · Ch = Chaitra.

Tier chips: **TOP /17** = topic asked in 7 or more of the 17 papers · **HOT /17** = 4–6 papers · **PIN /17** = 1–3 papers but worth ≥ 8 marks. **Mark chips** such as **2+5** are the marks that question actually carried. **verified/added** marks anything researched and added that was not in the source notes.